MANGALAYATAN UNIVERSITY

BESWAN ALIGARH, 202146

SESSION – 2025-26

मङ्गलायतन विश्वविद्यालय

॥ विश्वं ज्ञाने प्रतिष्ठितम् ॥

ASSIGNMENT

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Enrollment No… ; 202517735

Subject Name ; Engineering Mechanics

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Class ; Diploma

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Submitted To

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Sub:- Engineering Machanics

Sub. code :- DME - 201

Assignment - I

Q1) Explain the term force, write down its unit in SI, MKS & CGS System.

Ans

Force:

:- A Force is a push or a pull that can change the Shape, Size, direction, Speed or State of rest / motion of an object.

If Something moves, stops, changes directions, or deforms, a force is involved.

Its units:

(i) SI Units - Newton (N)

(ii) MKS - Newton (N)

(iii) CGS - Dyne (dyn)

Q2) Explain the Principle of transmissibility of force and disscuss its Practical implications in rigid body mechanics.

Ans

Principle of Transmissibility of Forces:

:- It states "if a forces acts at any Point on a rigid body, it may also be considered to act at any other point on its line of action, Provided this point is rigidly connected with the body."

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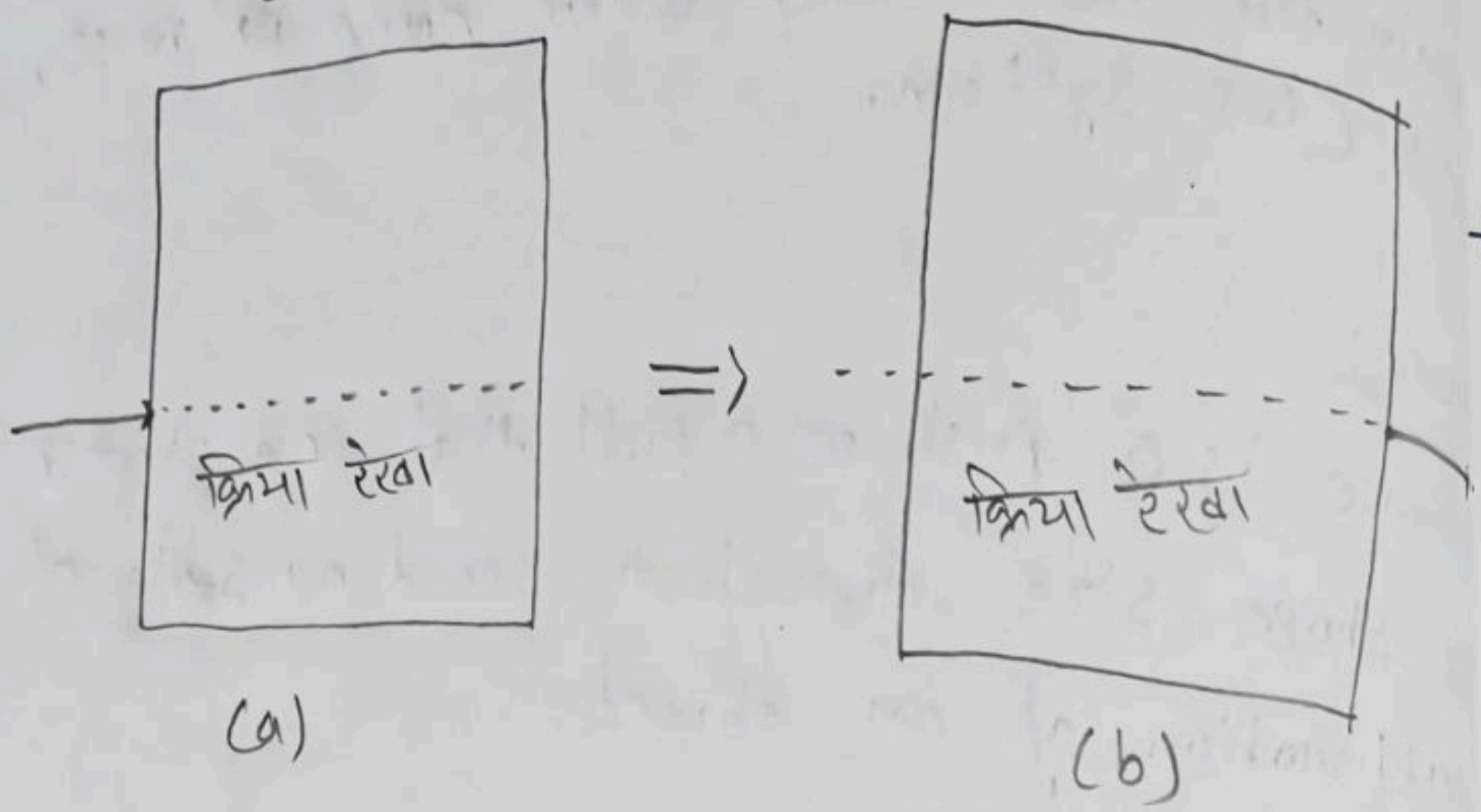

↳ means position along same line does not change motion of rigid body.

Examples:

Pulling a box with a rope - Pulling from front or middle gives same motion.

Practical Applications

1. Design of machines and structures
2. Beam analysis in civil ~~engni~~ engineering
3. Force distribution in mechanical parts
4. Use of cables and rods in bridges

Q3) Compare the triangle law, parallelogram law, and polygon law of forces with suitable illustrations.

Ans (a) Triangle Law of Forces:

:- If two forces acting on a body are represented by two sides of a triangle taken in order, the third side represents their resultant.

↳ used when only two forces act.

(b) Parallelogram Low of forces:

:- If two forces acting at a point are represented by two adjacent sides of a Parallelogram, the diagonal represents the resultant.

↳ Used to find magnitudes & direction of two forces.

(c) Polygon Law of Forces:

:- If many Forces acting on a body are represented by sides of a Polygon taken in order, the closing side gives the resultant.

↳ Used for multiple forces.

Comparison Table

	Law	Use	Forces Count
(a)	Triangle Law	Simple vector additions	02
(b)	Parallelogram Law	find magnitude & direction	02
(c)	Polygon Law	multiple forces	many

Q4) Define Engineering Mechanics. Explain its Significance and relevance in engineering applications with Suitable examples.

Ans Engineering Mechanics:

:- Engineering mechanics is the branch of Science that Studies forces and their effects on bodies.

It has two types (Parts):

(1) Statics - bodies at rest

(2) Dynamics - bodies in motion

Importance in Engineering

1. Design of buildings and bridges
2. Machine design
3. vehicle motion analysis
4. Robotics and automation

Examples:

- Bridge design uses Statics
- Car engine uses dynamics

↳ Note :- Engineering mechanics is the foundation of all engineerings branches.

Q5) Differentiate between scalar and vector quantities. Explains their importance in solving engineering problems.

Ans

Scalar Quantities:

:- The scalar quantities (or sometimes known as scalars) are those quantities which have magnitude only such as length, mass, time, distance, volume, density, temperature, speed. etc.

only magnitudes	No Direction

Vector Quantities:

:- The vector quantities (or sometimes known as vectors) are those quantities which have both magnitude and direction such as force, displacement velocity, acceleration, momentum etc.

magnitude + Direction

Difference Table:

Scalar Quantities	Vector Quantites
(i) only magnitude	(i) Magnitude + Direction
(ii) Simple addition	(ii) vector addition
(iii) Ex:- Mass, Time	(iii) Ex:- force

Importance in Engineering:

1. Correct calculations of forces.
2. Motion analysis
3. Structure Stability
4. Machine design

↳ Notes:- without vectors, engineering problems cannot be solved correctly.

Q6) Explain about rigid and flexible body with examples.

Ans

Rigid body:

:- A rigid body is defined as a definite quantity of matter, the parts of which are fixed in position relative to each other. Physical bodies are never absolutely but deform slightly under the action of loads. If the deformation is negligible as compared to its size, the body is termed as rigid.

Flexible body:

:- A flexible body is one that undergoes large deformations when subjected to forces.

- The shape and size changes significantly under load.
- Examples:
  - Ropes
  - Cables
  - Chains
  - Belts.